

Lassonde School of Engineering

EECS 2031

Unix LAB 2-Shell Scripting

Objective

To write simple shell scripts to practice the concepts learned during Week 1, 2, and 3.

All problems have the same weight. Do any 10. Save 10 separate scripts (files) and demo them to the TA.

Add the following header to each file:

```
# Author: Last name, first name
# Student ID: 123456789
# Login ID: EECS login ID
```

Problem 1: Write a shell script named `phone.sh` that prompts the user to enter a file name (which is a phone book in the current working directory) and a pattern (string). The script then searches the file and displays each line that matches the pattern entered. The pattern to be searched for is case-insensitive. Use the `phone_book` file as on eclass lab1 directory.

```
% phone.sh
Enter the file name: pbook.txt
Enter the pattern: alex
alex johnson      (416) 555-1234      family doctor
Alexander Smith  (905) 555-9876      home renovation contractor
```

Problem 2: Repeat Problem 1, except that the file name and the pattern to search for are now entered as command line arguments (parameters). Name the script `phoneArg.sh`.

```
% phoneArg.sh pbook.txt alex
alex johnson      (416) 555-1234      family doctor
Alexander Smith  (905) 555-9876      home renovation contractor
```

Problem 3: Write a script called `setRead.sh` that accepts a file name as a command line argument (parameter). It then sets the appropriate permission to allow the file to be read by everyone, displays a confirmation message, and lists the file with all the details for verification. Assume that the file exists and is valid so that no error checking is required.

```
% setRead.sh enum.c
File enum.c now readable by everyone
-rw-r--r-- 1 vashistp staff 436 1 May 16:19 enum.c
```

Problem 4: Write a shell script name `calc.sh` to implement a simple calculator that accepts input in the following format and displays the result of the calculation:

```
calc [operand_1] [operator] [operand_2]
```

The operands `operand_1` and `operand_2` are integers. The operator is one of the following: addition (+), subtraction (-), multiplication (x), division (/) and modulo (%).

Note: For the multiplication operator in the command line arguments, use letter 'x'. If you use the asterisk '*', your program will not work properly. Assume that all command line arguments are valid, and no error checking is required.

```
% calc 200 + 100
300
% calc 100 - 450
-350
% calc -50 x 50
-2500
% calc 30 / 7
4
% calc 29 % 12
5
```

Problem 5: Write a shell script named `exitCode.sh` that accepts two command line arguments (parameters):

1. one file name that is valid (so that there is no error when it is listed using the "ls" command);
2. one file name that is not valid, e.g., the file does not exist.

List each file using the "ls -l" command. After each "ls" command is executed, display the exit code of the "ls" command.

```
% exitCode.sh sizeof.c alien.txt
-rw----- 1 vashistp faculty 174 Sep 18 11:45 sizeof.c
Exit code is 0
ls: cannot access alien.txt: No such file or directory
Exit code is 2
```

Problem 6: Write a shell script named `mult.sh` that accepts an integer N as a command line argument. Display the multiplication table of N x k, where k = 1, 2, ..., 10. *Hint:* Use a *while* loop.

```
% mult.sh 8
8 x 1 = 8
8 x 2 = 16
8 x 3 = 24
8 x 4 = 32
8 x 5 = 40
8 x 6 = 48
```

8 x 7 = 56
8 x 8 = 64
8 x 9 = 72
8 x 10 = 80

Problem 7: Write a shell script named [details.sh](#) that accepts a file name or directory name as a command line argument, then display the answers to the following questions.

- Does the file or directory exist or not? (If the file or directory does not exist then display an error message as shown below and exits.)
- Is it a directory or a regular file? (Assume that if it is not a directory than it is a regular file.)
- If it is a regular file, is it empty or not?
- Is it readable or not?
- Is it writable or not?
- Is it executable or not?

Following are some examples.

```
% details.sh myScript
The file or directory "myScript" exists.
It is a regular file.
It is not empty.
It is readable.
It is writable
It is not executable.
% details.sh Labs
The file or directory "Labs" exists.
It is a directory.
It is readable.
It is not writable
It is executable.
% details.sh ufo.txt
The file or directory "ufo.txt" does not exist.
```

Problem 8: Write a shell script named [manyCmds.sh](#) in which define a variable that holds the following commands: whoami, groups, pwd, ps, date, cal, in that order. Use the variable and a for loop to execute the above commands. Connect to an EECS server (e.g., red.eecs.yorku.ca) and log into your EECS account to run the script on the server. Capture a screen shot of the output and submit the screenshot together with the script. Following is a sample output.

```
crimson 371 % manyCmds.sh whoami groups pwd ps date cal
akeem19
ugrad guac_edu guac_ea guac_rl guac_alt hc_prism hc_public
/cs/home/akeem19/eecs2031/uLab4
    PID TTY          TIME CMD
3191403 pts/2    00:00:00 tcsh
3226541 pts/2    00:00:00 manyCmds.sh
3226544 pts/2    00:00:00 ps
Tue Mar 28 02:30:57 EDT 2023
    March 2023
Su Mo Tu We Th Fr Sa
      1  2  3  4
 5  6  7  8  9 10 11
12 13 14 15 16 17 18
19 20 21 22 23 24 25
26 27 28 29 30 31

crimson 372 %
```

Problem 9: Write a script named [setRead2.sh](#) that accepts several file names as command line arguments (parameters). It then sets the appropriate permission to allow each file that exists to be read by everyone, displays a confirmation message, and lists the file with all the details for verification. If a file does not exist, display an error message as shown below.

```
% setRead2.sh sizeof.c alien.sh ptr1.c
File "sizeof.c" now readable by everyone.
-rw-r--r-- 1 vashistp faculty 174 Sep 18 11:45 sizeof.c
File "alien.sh" does not exist.
File "ptr1.c" now readable by everyone.
-rw-r--r-- 1 vashistp faculty 558 May 9 15:14 ptr1.c
```

Problem 10: Write a script named [phoneArg2.sh](#) as specified in this lab for **Problem 1** with the following addition. The user can now specify multiple patterns (words) as command line arguments (parameters). *Hint:* Use the "shift" command.

```
% phoneArg.sh pbook.txt alex friend tim
alex johnson (416) 555-1234 family doctor
Alexander Smith (905) 555-9876 home renovation contractor
Ray Szeto (416) 564-6786 friend
Nikola Sandic (647) 209-7932 friend from work
Karim Khalil (905) 346-6892 friend
ROB Oxley (416) 723-8759 friend from university
Tim Yu (416) 570-9522 Jill's coach
Timothy Hawn (647) 761-8906 hair dresser
```

Problem 11: Write a script named `chgmode.sh` that accepts the following command line arguments (parameters):

- the first parameter is one of the three permissions: 'read', 'write', 'execute'.
- the remaining parameters are file names.

Depending on the permission specified, use the "chmod" command to give all users the permission to perform the operation (read, write, or execute) on all the files listed as the parameters. After each "chmod", display a confirmation message and list the file with all the details using "ls -l" for verification. *Hint:* Use the "shift" command.

```
% chgmode.sh execute prdir prtline countx showpid
File 'prdir' is now executable by all.
-rwx--x--x 1 vashistp faculty 122 Nov 23 2010 prdir
File 'prtline' is now executable by all.
-rwx--x--x 1 vashistp faculty 132 Mar 21 2020 prtline
File 'countx' is now executable by all.
-rwx--x--x 1 vashistp faculty 190 Mar 21 2020 countx
File 'showpid' is now executable by all.
-rwx--x--x 1 vashistp faculty 79 Nov 20 16:43 showpid
```

Deliverable

Demo to 10 problems solutions to your lab TA

Rubrics: Lab2 is for 10 marks. Students should be able to demo 10 tasks from 11 given above for maximum marks.

Please note that you need to create your own files and directories to be used in the tasks above. Only phone_book.txt has been provided on eClass (refer the file in lab1 directory on eclass).

Note: Try to complete and demonstrate lab2 in this week. If you cannot complete it, attempt most of it, and rest can be completed in the make up labs and demonstrated after completion. Keep in mind, make up labs session will also have sample questions to be practiced for labtest1, however, those questions are not graded and no demo is required.